Date: 12/10/26 ITA0328 Keerthika S

Day: Monday Mobile Computing 192371013

Col-ATol

slot A

8)

Concept map

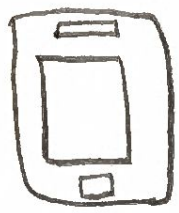
Public Land Mobile Network (PLMN)

Provides mobile communication services (Voice, sms, data, internet)



Operators

- Manage the network
- Provide services
- Maintain infrastructure
- Ensure network Performance



Subscribers

- Use mobile devices
- Access voice, SMS and internet
- Move across different locations



Switching Centers

- Route calls and data
- Connect different users
- Manage communication between networks



Radio access network [RAN]

- Provides wireless access
- Connects devices to the network
- Use base stations and radio signals.

How they Provide Seamless Connectivity

1. Operators control and manage network
2. Subscribers initiate Communication using mobile devices
3. RAN Provides wireless Connectivity between users and network
4. Switching Centers route calls and data to the correct destination

Justification of relationships

Operators ↔ Subscribers

Operators ↔ RAN

RAN ↔ Switching Centers

Switching Centers ↔ Subscribers

If any component fails, network Performance may decrease.



Coordination and Communication

All Components work together through standards and Protocols



Seamless Connectivity

Voice Calls • SMS • Internet • mobility

Enables continuous communication even when users move between different cell areas.



Operational Efficiency

High Coverage • Low delay • Reliability • Better Service quality